

WASP-36b

R-0693

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_260122 · created 2026-07-27T20:52:28+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461062.8542 +/- 0.0063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.055 +/- 0.048
Transit depth (Rp/Rs)^2	0.31 +/- 0.53 [%]
Orbital Inclination (inc)	80.83 +/- 2.93
Airmass coefficient 1 (a1)	77.96 +/- 47.62
Airmass coefficient 2 (a2)	-3.0 +/- 0.86
Scatter in the residuals of the lightcurve fit is	45.63 %
Best Comparison Star	None
Optimal Aperture	7.6
Optimal Annulus	8.24
Transit Duration (day)	0.038 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.055 ± 0.048 vs 0.1368 (deviation 1.3σ)
Duration fit vs published	0.038 d vs 0.0758 d (deviation 0.93σ)
Mid-time O–C	13.0 min
Depth SNR	0.9
Residual scatter	45.63 %

Light curve

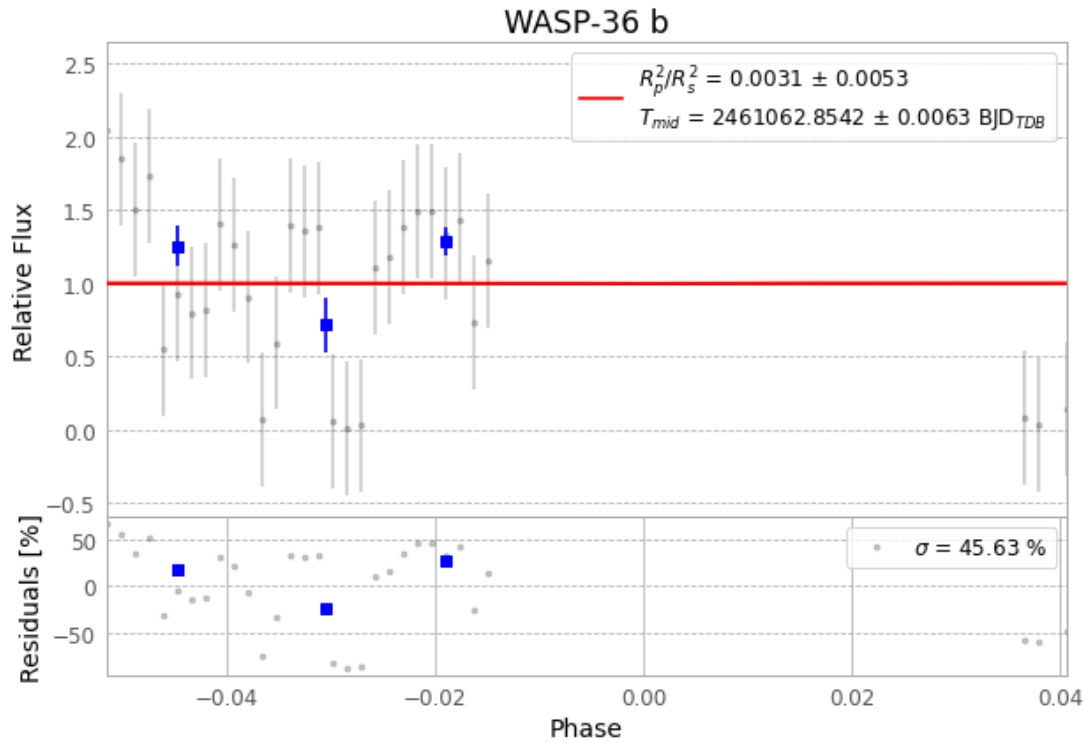


Figure 1 — FinalLightCurve_WASP-36 b_2026-01-21.png (SHA-256 39dfc312cec04f35...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [592, 466], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a6eb2f7bc5f48633...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

73 FITS frames (48 MB), WASP-36260122062716.FITS → WASP-36260122100315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-260122.log (0ffc9dbcd766...), FinalLightCurve_WASP-36 b_2026-01-21.png (39dfc312cec0...), FinalLightCurve_WASP-36 b_2026-01-21.csv (883a1cbbb746...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 20:52Z.